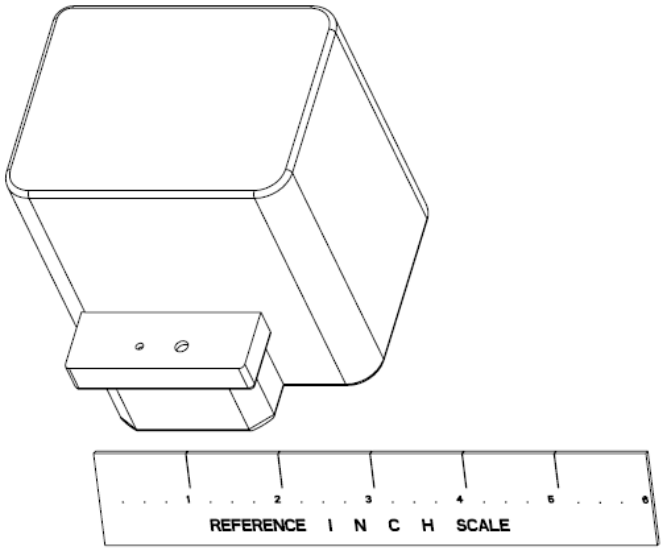
INCH-POUND
MIL-PRF-32271/14
30 October 2008

PERFORMANCE SPECIFICATION SHEET

BATTERY, NON-RECHARGEABLE, LITHIUM

This specification is approved for use by all Departments
and Agencies of the Department of Defense.

The requirements for acquiring the product herein shall consist of this specification and MIL-PRF-32271.



See NOTES for nominal dimensions; see applicable drawing for dimensional requirements

Figure 1 – General View, M32271/14 Battery Shape

REQUIREMENTS (see 1.2 for Type, Class, and Features descriptions):

Type – II
Class – 2
Features – C
PIN – M32271/14-22C
Approved chemistry: Lithium sulfur dioxide (Li/SO_2)

Specification requirements: The following requirements of MIL-PRF-32271, identified therein as "when specified", are applicable as indicated below:

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Requirement	Specification Reference	Applicability
Parallel cell arrangements	3.4	No
Connectors	3.4.5	Yes
Battery charger connection	3.4.5.1	No
Connection integrity	3.4.5.2	No
Terminal integrity	3.4.6	No
Complete discharge device	3.4.7	Yes
State of charge device	3.4.8	No
State of charge data output	3.4.8e	No
Cell charging	3.5.3	Yes
Nail penetration	3.5.4	No
Cell series string short circuit	3.5.6	No
Parallel string charge protection	3.5.6.1	No
Charge protection	3.5.11	Yes
Over-current protection	3.5.13.1	Yes
Over-temperature protection	3.5.14	Yes
Surface temperature	3.6h	Yes
Capacity tests LR and LRT		Yes
Immersion	3.7.8	Yes
Watertight integrity	3.7.9	No
DO NOT CHARGE! marking	3.8.1.2b	No
Complete discharge device marking/label	3.8.4	Yes
State of charge marking	3.8.5	No

Dimensions, marking and configuration

Battery – Drawing A3315891
Battery Connector (3.4.5 & 4.6.1.7) – Drawing A3315891
Mating Connector (3.4.5 & 4.6.1.7) – Drawing A3315891
Battery Charger Connector – Not Applicable

Battery voltages (3.4.4 & 4.6.1.5):

Battery open-circuit voltage (3.4.4.1 & 4.6.1.5.1):
Maximum – 27.45 volts
Minimum – 26.00 volts

Battery closed circuit voltage (3.4.4.2 & 4.6.1.5.2):
For capacity, initial voltage delay, and closed circuit voltage test:
Minimum – 18.0 volts

Cell closed circuit voltage test (3.4.4.3 & 4.6.1.3): Minimum - 2.0 volts per cell

Maximum weight (3.1 & 4.6.1.6): 1.43 lbs. (649 g)

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Capacity test requirements (3.6 & 4.6.4):

Test	Initial voltage delay Time (MAX) to	Volts	Minimum Capacity		Cut-off Volts
				Storage Period	
			1-Week	4-Week	
I	5 seconds	18.0	3.4 hours		18.0
L	60 seconds	18.0	1.7 hours		18.0
H	5 seconds	18.0	6.2 hours		18.0
LR	5 seconds	18.0	49.0 hours		3.0
IT	5 seconds	18.0	3.2 hours	3.0 hours	18.0
LT	60 seconds	18.0	1.6 hours	1.5 hours	18.0
HT	5 seconds	18.0	5.7 hours	4.7 hours	18.0
LRT	5 seconds	18.0	45.6 hours	44.1 hours	3.0

Abuse test pulse discharge minimum capacity requirement (3.5.12 & 4.6.2.12i): 3.2 ampere-hours

METHODS OF EXAMINATION AND TEST:

Verification requirements. The following verification requirements of MIL-PRF-32271, identified therein by the phrase “when specified,” are applicable as indicated below:

Test Requirement	Specification Reference	Applicability
Parallel discharges	Tables III, IV, IX, & XI	No
Connector	4.6.1.7	Yes
Battery charger connection test	4.6.1.7.1	No
Static connection integrity	4.6.1.7.2	No
Dynamic connection integrity	4.6.1.7.3	No
Terminal integrity	4.6.1.8	No
Complete discharge device	4.6.1.9	Yes
State of charge device	4.6.1.10	No
Cell charging	4.6.2.3	Yes
Nail penetration	4.6.2.4	No
Cell series string short	4.6.2.6	No
Parallel cell charge protection	4.6.2.6.1	No
Charge protection test	4.6.2.11	Yes
Battery over-current protection	4.6.2.13	Yes
Battery over-temperature protection	4.6.2.14	Yes
Surface temperature	4.6.4.1	Yes
Capacity test LR	4.6.4.1.5	Yes
Capacity test LRT	4.6.4.1.9	Yes
Immersion	4.6.5.8	Yes
Watertight integrity	4.6.5.9	No

Cell closed-circuit voltage test: When cells are tested as specified in 4.6.1.3, load each cell with 1 ampere constant current, 2 ohms resistance, or 2 watts constant power.

Battery closed-circuit voltage test: When tested as specified in 4.6.1.5.2, load each battery with 1 ampere constant current, 18 ohms resistance, or 18 watts of power.

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Abuse test, pre-discharge: When tested as specified in 4.6.2.12a, discharge with a load of 1.0 ampere for 1.7 hours.

Abuse test, pulse discharge: When tested as specified in 4.6.2.12i, discharge with a load of 1.4 amperes for 1 minute followed by 0.9 amperes for 4 minutes, cycled continuously to 18.0 volts.

Battery over-current protection: When tested as specified in 4.6.2.13, load batteries with either 2.0 amperes or 36 watts.

Immersion: When tested as specified in 4.6.5.8, apply a load of 425 milli-amperes during the storage and immersion time periods specified.

Capacity tests (4.6.4):

Test	Discharge Rate	Duty Cycle
I	1.0 ampere	Continuous discharge to cut-off volts, followed by 1.0 ampere To zero volts, then 1.0 ampere forced discharge for 10 minutes.
L	1.0 ampere	Continuous discharge to cut-off voltage
H	1.0 ampere	Discharge for 5 minutes at 1.0 ampere, followed by 5 minutes on open circuit; repeat cycle continuously to cut-off voltage
LR	75 mA/2.5W	Discharge continuously at 75 milli-amperes to 18.5 volts followed By continuous discharge at 2.5 watts to cut-off voltage
IT	1.0 ampere	Continuous discharge to cut-off voltage
LT	1.0 ampere	Continuous discharge to cut-off voltage
HT	1.0 ampere	Discharge for 5 minutes at 1.0 ampere, followed by 5 minutes on open circuit; repeat cycle continuously to cut-off voltage
LRT	75 mA/2.5W	Discharge continuously at 75 milli-amperes to 18.5 volts followed By continuous discharge at 2.5 watts to cut-off voltage

NOTES:

(This section contains information of a general nature which may be helpful, but is not mandatory)

Intended use: This battery is intended for use in one model of respiratory protection for Army helicopter pilots.

[1]Nominal ratings: The following are the nominal ratings for the battery described by this specification sheet. They are provided for information purposes.

[1] Capacity and energy delivered by a battery are significantly affected by usage conditions, such as temperature and loads applied. If you have any questions about use of this battery in a particular device or circumstance, please visit the following web site (contact info is posted): http://www.cerdec.army.mil/c2d/armypower.

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Battery PIN:	**M32271/14-12E (Li/MnO_2)**
Overall Dimensions::	3.345 in. H x 3.39 in. W x 3.34 in. D
MAX Weight:	1.43 lbs (649 grams)
Voltage Range:	18.0-27.45
Nominal Capacity: (in ampere-hours)	3.4 Amp-hrs
Nominal Energy: (in watt-hours)	77 Watt-hrs
MAX rated power output:	23 watts
MAX continuous load rating:	1.0 ampere
MAX pulse load rating:	1.4 amperes
Instantaneous trip rating:	2.0 amperes
Operating temperature range:	-20 to 130ºF (-29 to 55ºC)
Storage temperature range:	-40 to 160ºF (-40 to 71ºC)
MAX abusive temperature: (non-operating)	195ºF (91ºC)

Other data:

Military Type Designations. The military type designation that relates to the PIN covered by this specification sheet is as follows: the BA-5093()/U designation has been used for the Type II PIN.

This battery is subject to the transportation requirements of 49 CFR 173.185. The battery has a liquid cathode and more than 1.0 grams of lithium content. See applicable Material Safety Data Sheet (MSDS) for the maximum lithium weight per cell and battery. It should be noted that all non-rechargeable lithium batteries are restricted from shipment as cargo aboard passenger aircraft within, entering, or leaving the US.

This battery includes a complete discharge device in order to render the battery non-reactive after use. If a battery is damaged, or if the device does not operate as indicated by an amber or yellow light, the battery is considered reactive waste. This battery may be recycled after use. Universal waste rules, where applicable, apply only when recycling is the chosen disposal method.

The US Army CECOM Life Cycle Management Command (LCMC) publishes guidelines for the design of battery compartments for devices using the battery covered by this specification sheet: Technical Bulletin CECOM-TB-7. Please contact the CECOM LCMC Directorate for Safety at monm-amselsfsec@conus.army.mil for further information.

Navy safety tests of NAVSEA S9310-AQ-SAF-010 are required for this battery during first article testing. The test data provided from the testing will be used to evaluate applications for specific using devices and Navy platforms. Devices using this battery will require US Navy Safety Approval prior to use by Department of Navy users. Please consult with NAVSEA Instruction 9310.1 for further information.

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Army – CR
Navy – NW
Air Force – 99
DLA – GS

Review activities:
Navy – SH, AS, MC
Air Force – 71

Preparing activity:
Army – CR
(Project Number 6135-2006-017)

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